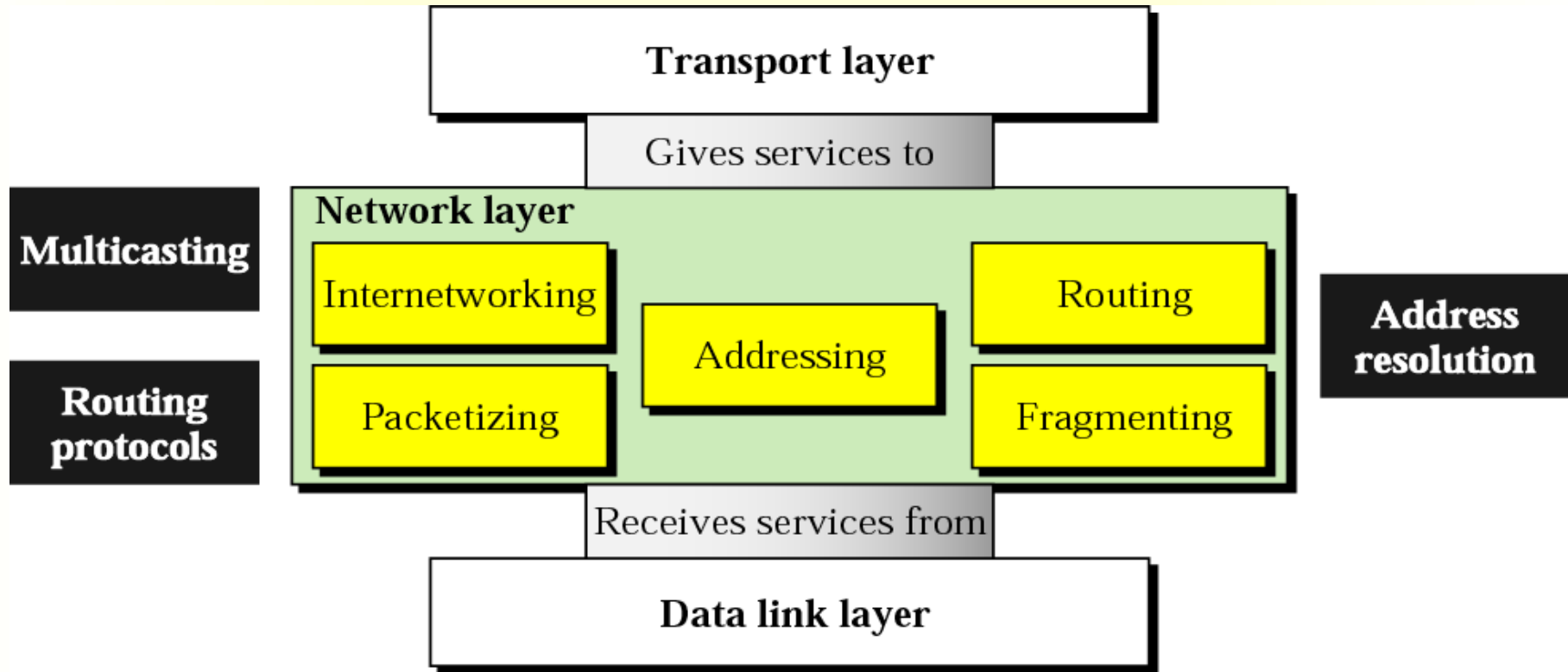


Chapter 4.

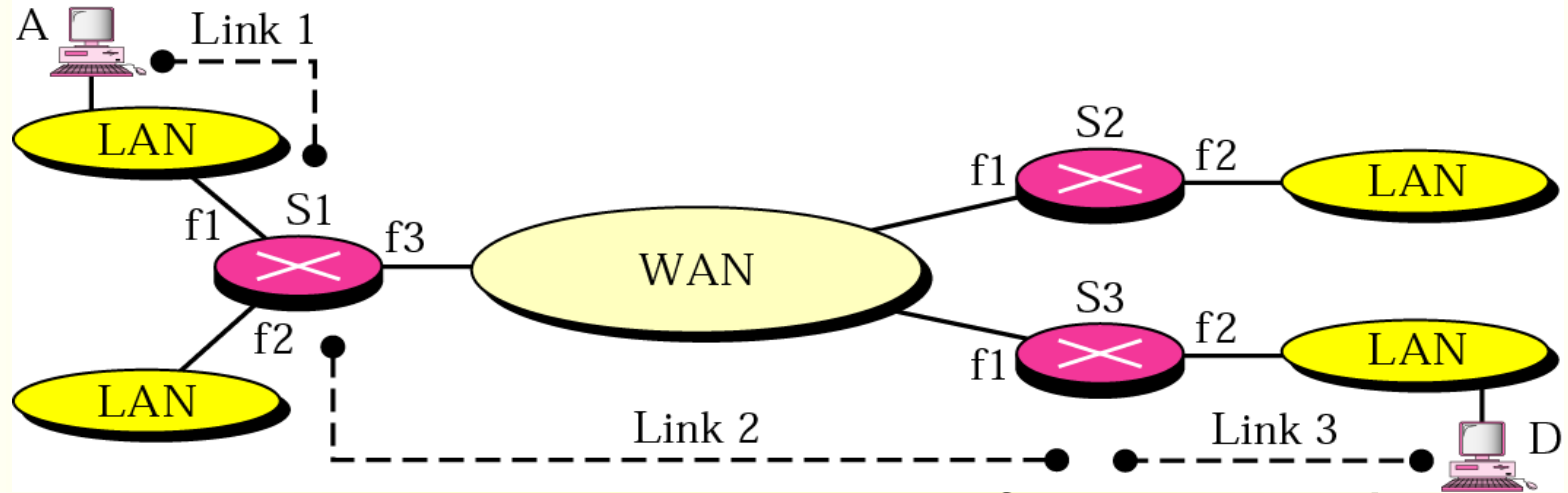
The Network Layer

- the network layer is responsible for carrying a packet from one computer to another; it is responsible for **host-to-host** delivery
- position of the network layer



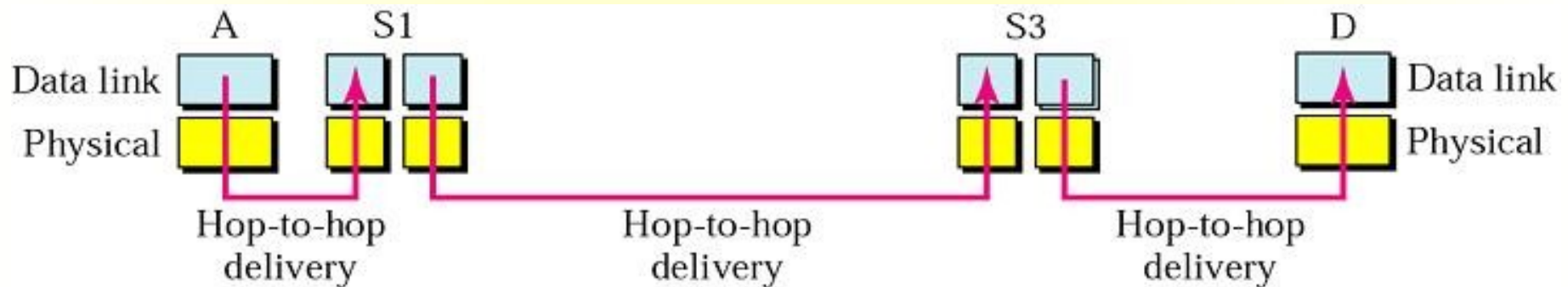
4.1 Internetworking

- internetworking refers to the logical gluing of **heterogeneous** physical networks together to look like a single network to the upper transport and application layers



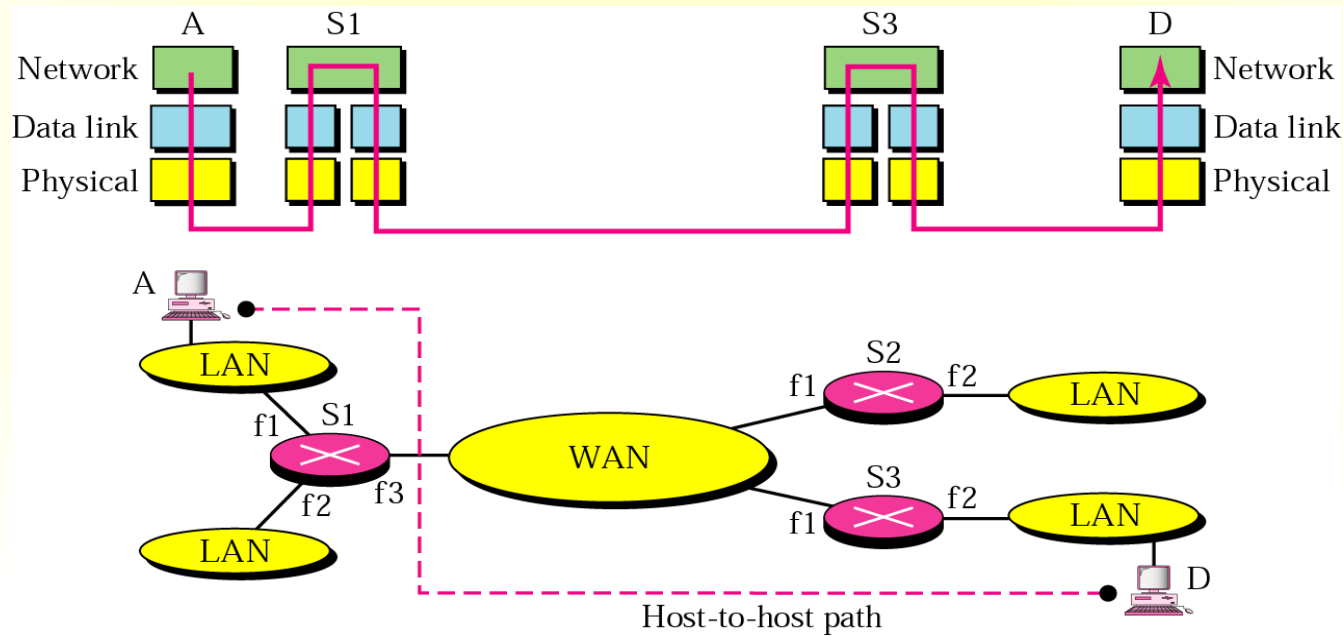
- the above internetwork is made up of 5 networks: 4 LANs and 1 WAN
- S_i : switch or router; f_i : interface
- data sent from A to D passes through 3 links
- nomenclature:
 - the source and the destinations are usually referred to as **hosts**
 - a host or a router is referred to as a **hop**

- in each link, 2 physical and data link layers are involved
- problem:
 - when data arrives at S_1 , how does it know that it should be sent to S_3 ?
 - the physical and data link layers do not help S_1
 - the frame does not carry any routing information, it contains the MAC addresses of A as the source and that of S_1 as the destination (for a LAN or a WAN, delivery means carrying the frame through one link, not beyond)



- the network layer is responsible for host-to-host delivery and for routing the packets through the routers or switches

- **network layer at source**
 - receives data from the transport layer
 - adds the **universal** addresses of A and D
 - makes sure the packet is of correct size for passage through the next link, fragment it if necessary
- **network layer at router or switch**
 - responsible for routing the packet using a routing table
 - the packet may go through another fragmentation if need be
- **network layer at destination**
 - address verification; error detection; reassembling the fragments
 - delivering to the transport layer



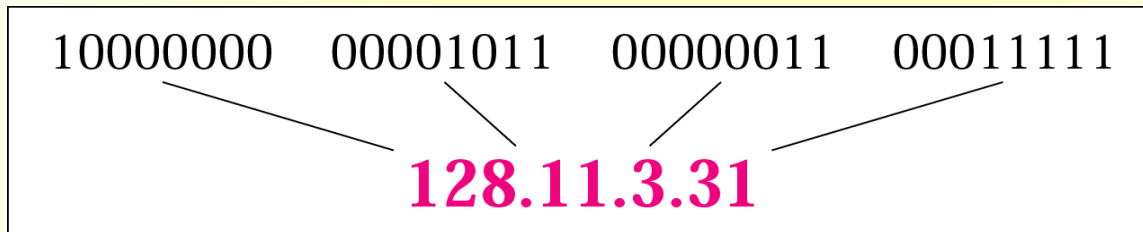
4.2 Addressing

- to **uniquely** and **universally** identify each device on the Internet to allow global communication between all devices
- analogous to the telephone system; the Department of Computer Science: 251 011 1222922, wherever you are on the globe
- each address belongs to a single host, but a single host can have multiple addresses (if it has multiple connections to the Internet)
- the identifier used in the network layer of the Internet model is called the **Internet address** or **IP address**
- it is a 32-bit binary address (in IPv4)

- two common notations: binary notation and dotted-decimal notation
 - **binary notation**: one or more spaces inserted between each octet (8 bits)

10000000 00001011 00000011 00011111

- **dotted-decimal notation**: to make it more compact and easier to read (for humans)

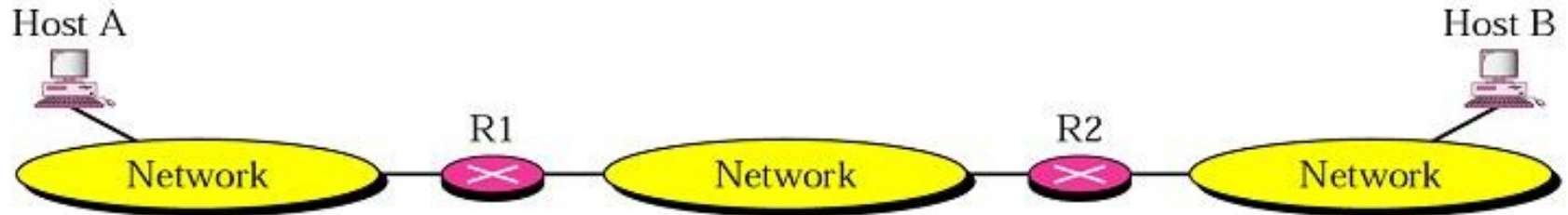


- we will study IP addresses later

4.3 Routing, Routing Protocols and Multicasting

- making a decision and choosing one route whenever there are multiple routes
- based on some criteria; how do you choose a route when you drive? may be the condition of your car and the road, the shortest one, the one that is not congested, the one with less traffic lights, avoid forbidden paths (American military packets may not pass through Russia), ...
- it is the job of the network layer **routing protocol**; a combination of rules and procedures that lets routers in the internet inform one another of changes (to share whatever they know about the internet or their neighborhood); e.g. a failure of a network can be communicated
- at the heart of such protocol is the **routing algorithm** that determines the path for a packet

- routing requires a host or a router to have a **routing table** which is constructed by the routing algorithm



Routing table for host A

Destination	Route
Host B	R1, R2, Host B

Routing table for R1

Destination	Route
Host B	R2, Host B

Routing table for R2

Destination	Route
Host B	Host B

- given big internetworks such as the Internet, the number of entries in the routing table becomes large and table look ups become inefficient; methods for reducing its size are required

- **next-hop routing**

- the routing table holds only the information that leads to the next hop

Routing table for host A

Destination	Next Hop
Host B	R1

Routing table for R1

Destination	Next Hop
Host B	R2

Routing table for R2

Destination	Next Hop
Host B	—

- **network-specific routing**

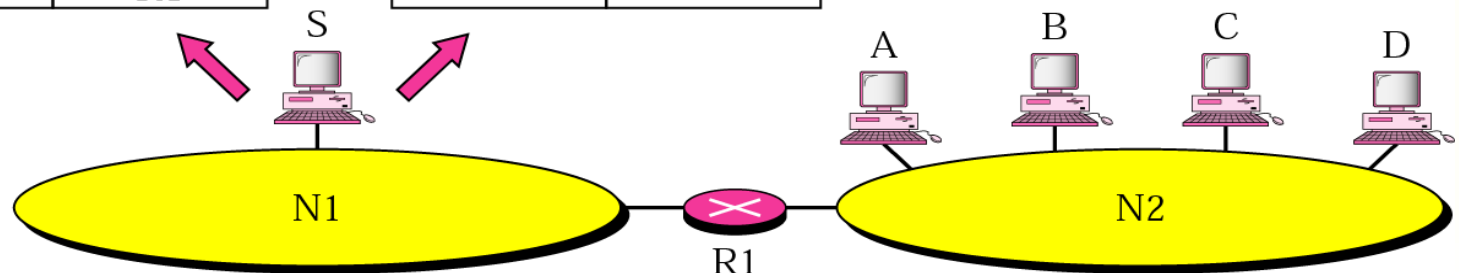
- instead of having an entry for every host connected to the same physical network, there is only one entry to define the address of the network itself

Routing table for host S based on host-specific routing

Destination	Next Hop
A	R1
B	R1
C	R1
D	R1

Routing table for host S based on network-specific routing

Destination	Next Hop
N2	R1

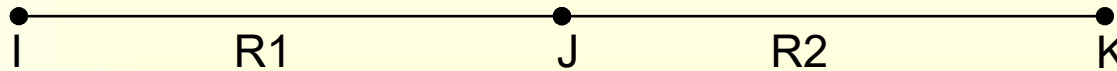


- two kinds of routing algorithms (routing tables):
nonadaptive (static) and **adaptive (dynamic)**
- **nonadaptive (static)**
 - routing decisions are not based on measurements or estimates of the current topology or traffic
 - the choice of a route is computed in advance, off-line, and downloaded to the routers when the network is booted **or**
 - an administrator enters the route for each destination into the table
 - not automatically updated when there is a change
 - may be used in a small internet, but not for big internet like the Internet

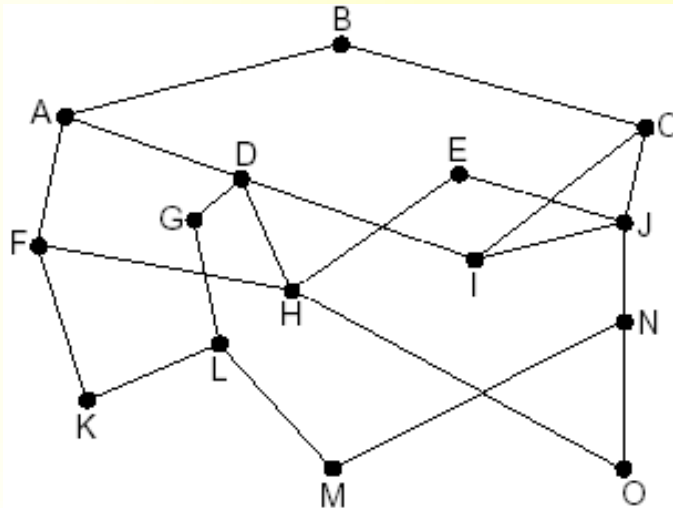
- **adaptive (dynamic)**
 - **routing decisions are made periodically (every δ sec) to reflect changes in the topology, traffic, a shutdown of a router, a break in the link, a better route has been created, ...**

- **interior** and **exterior** routing (protocols)
 - since an internet can be large, one routing protocol cannot handle the task of updating the routing tables of all routers
 - hence, an internet is divided into autonomous systems
 - an **autonomous system (AS)** is a group of networks and routers under the authority of a single administration
 - routing inside an autonomous system is referred to as **interior routing**; each AS can choose its own routing protocol
 - routing between autonomous systems is referred to as **exterior routing**; one protocol is usually chosen to handle routing between autonomous systems; usually used for routing in the Internet

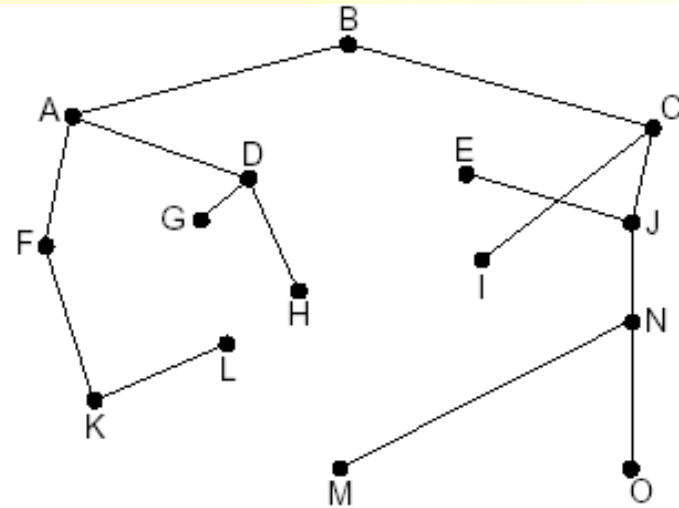
- the **optimality principle**: the optimal route
 - if router J is on the optimal path from I to K, then the optimal path from J to K also falls along the same route
 - why? if a route better than R2 existed from J to K, then the route from I to K could be improved by concatenating it with R1



- the consequence of the optimality principle is that “the set of optimal routes from all sources to a given destination form a tree rooted at the destination”
- such a tree is called a **sink tree**



a subnet



sink tree for router B

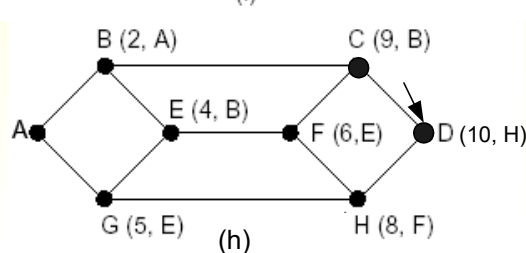
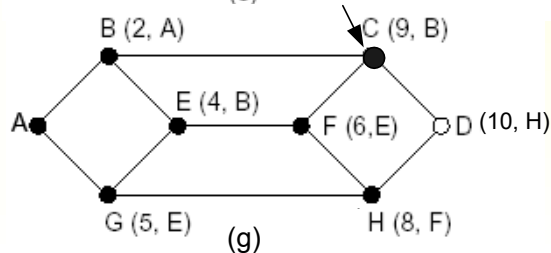
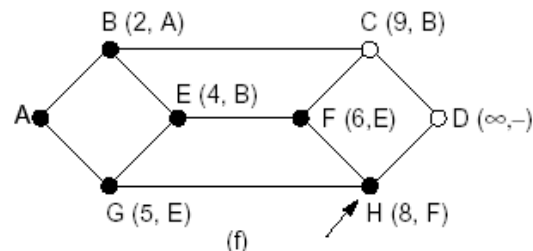
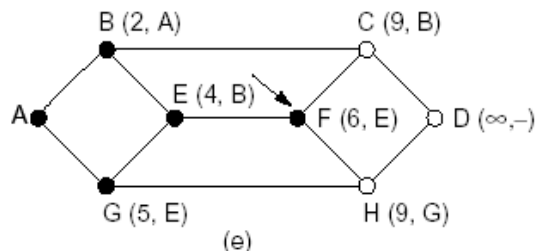
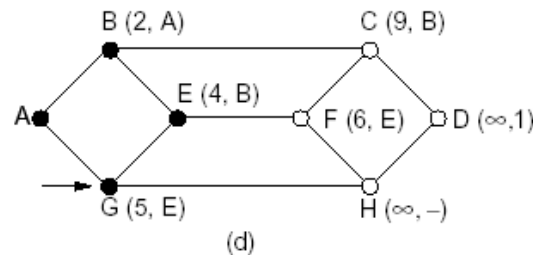
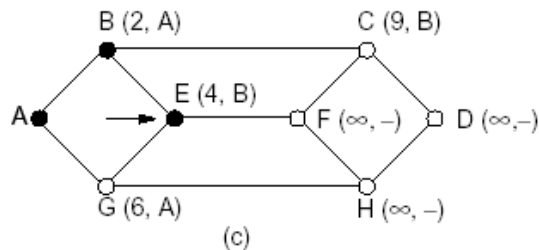
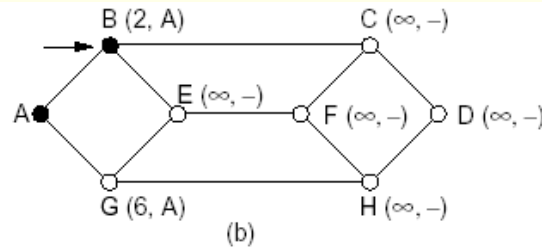
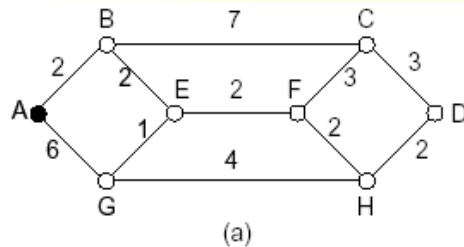
(distance metric is the number of hops)

- the sink tree is not necessarily unique; as a tree it does not contain any loops; $\Rightarrow$ each packet will be delivered within a finite and bounded number of hops; at least in theory
 - the goal of a routing algorithm is to discover the sink trees for all routers
 - the optimality principle and the sink tree serve as **benchmarks** to measure routing algorithms
- i. **Shortest Path Routing** (static) for unicast routing (by Dijkstra)
- aim: build a **graph** of the subnet - each node of the graph representing a router and each arc representing a link; the arcs are labeled as a function of any one of the metrics (distance, hop count, ...)
 - to select a route between two routers, the algorithm finds the “shortest” path between them on the graph

§ how does the algorithm work to find the shortest path?

§ each node is labeled with its distance from the source; initially labeled ∞ since no path is known

§ a label may be **tentative** or **permanent** (when the shortest



■ we want to find the path from A to D
→ working node

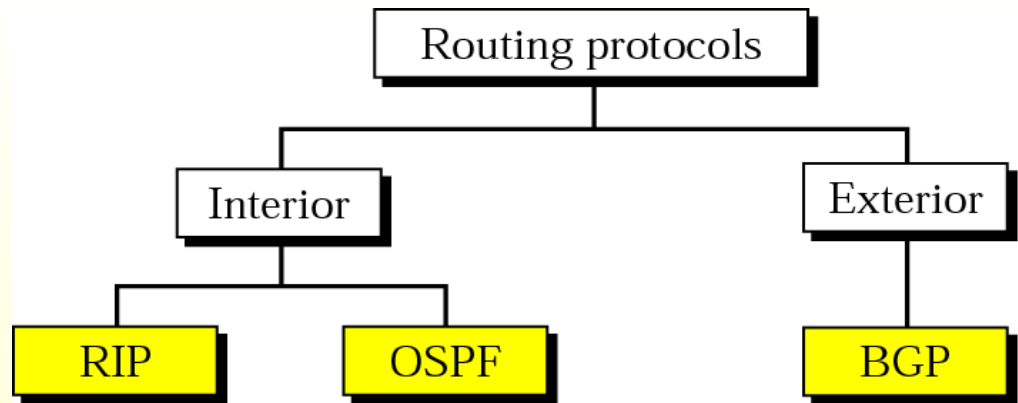
■ the shortest path is ABEFHD

ii. Flooding (static) for multicast and broadcast routing

- every incoming packet is sent out on every outgoing line except the one it arrived on
- packets received a second time are discarded
- it generates too many duplicate packets
- some measures required to avoid having an infinite number of packets; e.g. have a hop counter included in the header of each packet; initially initialized by the number of hops from source to destination; each hop decrements it and the packet is discarded when the counter is zero
- **selective flooding** is a variant; send only on those lines that are going approximately in the right direction

- possible applications include
 - to update all databases concurrently in a distributed database system (multicast routing)
 - in wireless networks where every station can listen (broadcast routing)
 - if there is a possibility of many routers being destroyed such as in a military application (may be for unicast transmission too)
- flooding is not **mostly** practical
- however, since flooding always chooses the shortest path, it can serve as a benchmark for other routing algorithms

Unicast Routing Protocols



- Interior

- i. RIP - Routing Information Protocol

- uses distance vector routing algorithm (dynamic)
 - also called distributed Bellman-Ford routing algorithm or Ford-Fulkerson algorithm
 - each router maintains a table (one entry per router) giving the best known distance to each destination

Destination	Hop Count	Next Router	Other information
163.5.0.0	7	172.6.23.4	
197.5.13.0	5	176.3.6.17	
189.45.0.0	4	200.5.1.6	
115.0.0.0	6	131.4.7.19	

- the tables are updated by exchanging information with neighbors
- in the general case, the metrics could be: number of hops, time delay in milliseconds, total number of packets queued along the path, ...
- it is the original ARPANET routing algorithm (using queue length as a metric) and was used until 1979
- the distance metric used in RIP is the number of hops
- ⇒ all networks are treated as equals, i.e., the cost of passing through each network is the same, it is one hop count

ii. OSPF - Open Shortest Path First

- became a standard in 1990 by the IETF (Internet Engineering Task Force)
- aim: to simplify the implementation at the boundaries between Autonomous Systems by having a standard for the interior routing
- IETF identified some requirements
 - has to be published in the **open** literature - proprietary solutions are not good
 - has to support a variety of distance metrics (physical distance, delay, ...)
 - it has to be dynamic
 - has to do load balancing for better performance (the best route is not always the best)

Network	Cost	Next Router	Other Information
N1	5	C	
N2	7	D	
N3	10	B	
N4	11	D	
N5	15	C	

- has to support routing based on type of service
- support for hierarchical systems - routers are not expected to know the entire topology - see later
- introduce security so that spoofing routers by sending false routing information is avoided
- uses **link state routing** algorithm (dynamic)
 - replaced distance vector routing in ARPANET
 - since distance vector routing does not consider bandwidth (no need initially since all lines were 56 Kbps; later some lines were upgraded) and other metrics
 - it uses Dijkstra's algorithm to find the shortest path and construct the routing table

■ Exterior

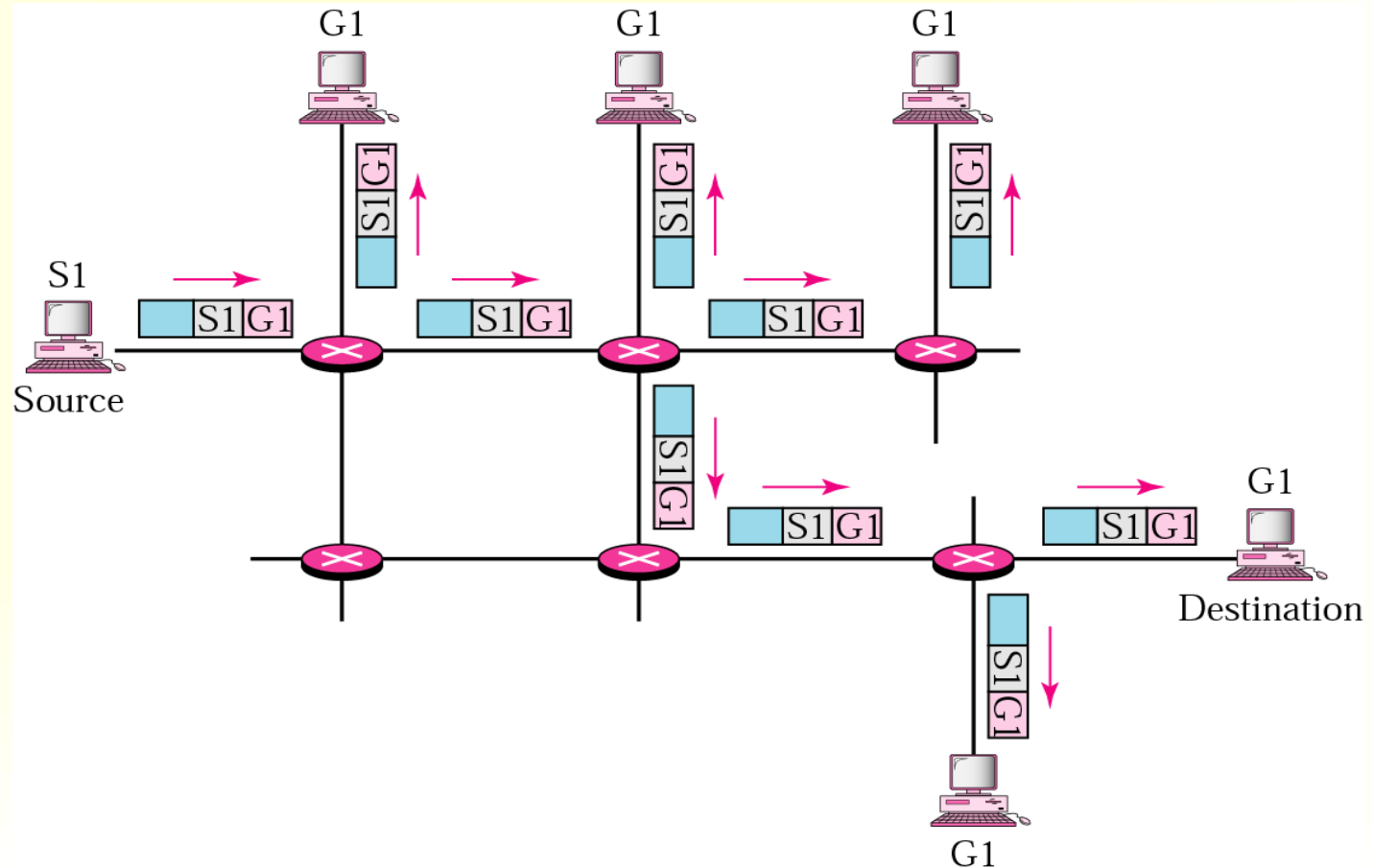
BGP - Border Gateway Protocol

- first appeared in 1989
- it uses **path vector routing** algorithm (dynamic)
- problems with the other algorithms
 - the route with the smallest hop count may not be the preferred route (if an autonomous system in which a packet passes is not secure, even though it is the shortest and for other political reasons)
 - they only announce the hop counts without actually defining the path that leads to the destination
- in path vector routing, each entry in the routing table contains the destination network, the next router, and the path to reach the destination

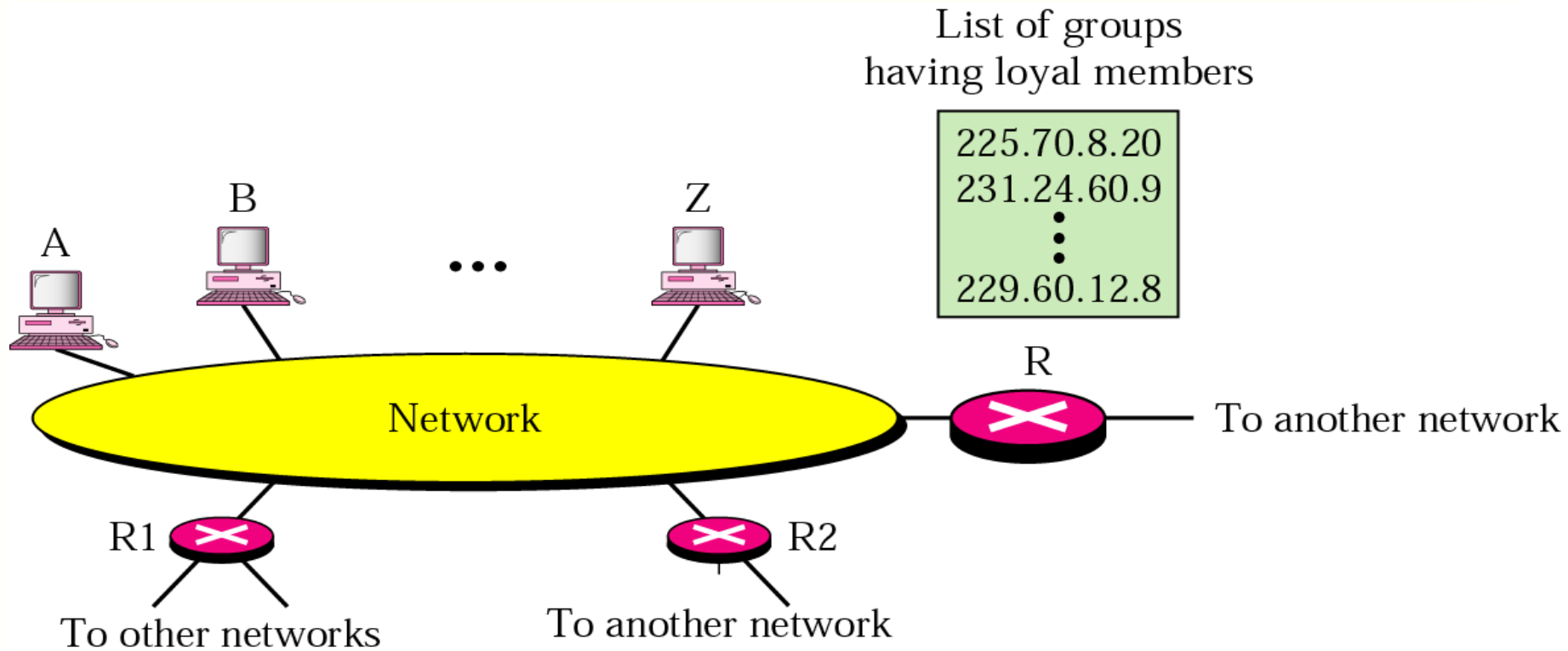
Network	Next Router	Path
N01	R01	AS14, AS23, AS67
N02	R05	AS22, AS67, AS05, AS89
N03	R06	AS67, AS89, AS09, AS34
N04	R12	AS62, AS02, AS09

■ Multicast Routing

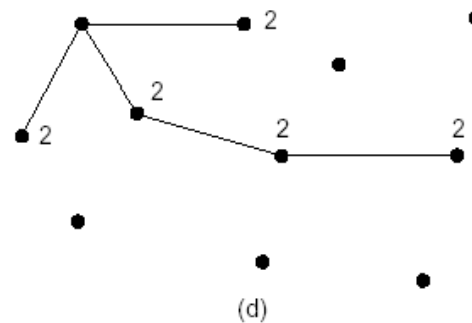
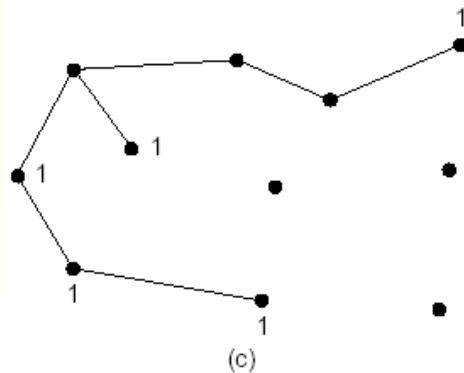
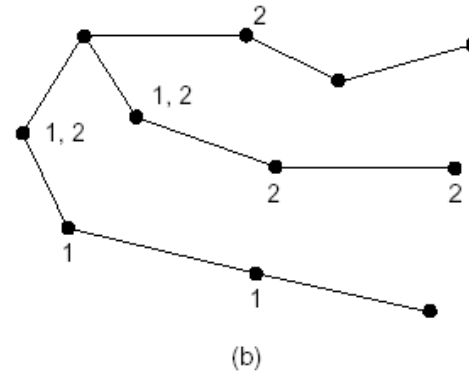
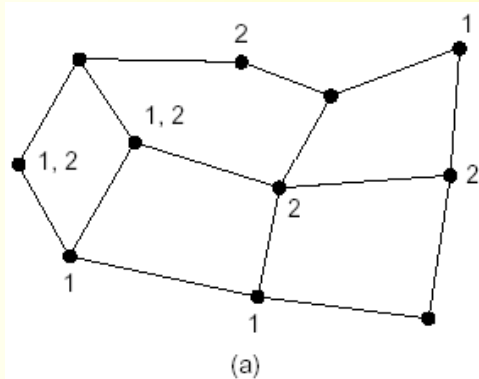
- multicasting: delivery of data from one host to many destinations; as a result of multimedia applications
- a one-to-many relationship
- a multicast router may forward a received packet through several of its ports (or discard it if it is not in the multicast path)



- it requires group membership management - to create and destroy groups, to allow processes (loyal members) to join and leave groups
- one such protocol is **IGMP** (Internet Group Management Protocol)
- has various versions - we don't deal with the details
- not a concern of the routing algorithm; routers should only know which of their hosts belong to which groups
- a multicast router connected to a network has a list of multicast addresses of the groups for which the router distributes packets to groups with at least one **loyal** member in that network
- for each group, there is one router which has the duty of distributing the multicast packets destined for that group



- for multicasting, each router computes a spanning tree covering all other routes
 - a. entire network with two groups where a node may belong to a single group, two groups, or none (see the numbers on the nodes)
 - b. a spanning tree for the left most router
 - c. pruned spanning tree for Group 1
 - d. pruned spanning tree for Group 2



- several ways of pruning the tree exist
- the simple one is if link state routing is used - each router is aware of the complete topology, including which hosts belong to which groups

Multicast Routing Protocols

- there are several multicast routing protocols (implemented or proposed)
- **DVMRP - Distance Vector Multicast Routing Protocol**
 - an extension of distance vector routing
- **MOSPF - Multicast Open Shortest Path First**
 - an extension of the OSPF protocol
- **CBT - Core-Based Tree**
 - it uses a core (center router or rendezvous router) as the root of the tree
 - an autonomous system is divided into regions, and a core is chosen for each region

- **PIM - Protocol Independent Multicast**
 - it has two varieties
 - **PIM-DM - Protocol Independent Multicast-Dense Mode**
 - used when there is a possibility that each router is involved in multicasting
 - **PIM-SM - Protocol Independent Multicast-Sparse Mode**
 - used when there is a slight possibility that each router is involved in multicasting
- multicasting has many applications
 - access to distributed systems
 - information dissemination
 - distance learning
 - multimedia communications

■ **Broadcast Routing Protocols**

- **sending a packet to all destinations simultaneously**
- **possible applications**
 - **distributing weather reports, stock market updates, live radio programmes**
- **the Internet does not explicitly support broadcasting (creates a huge amount of traffic and needs a huge bandwidth)**
- **methods**
 - 1. send the packet to each destination**
 - **wastes bandwidth**
 - **the source must have a complete list of all the destinations - very big table**
 - 2. flooding**
 - **generates too many packets and consumes too much bandwidth**

3. multidestination routing

- each packet contains either a list of destinations or a bit map - major problem
- a router receiving a packet generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line
- after some hops, each packet will carry only one destination and can be treated as a normal packet

4. using a spanning tree

- used by the originating router and the subsequent ones
- good use of bandwidth
- but each router must have a knowledge of some spanning tree

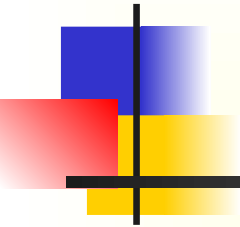
5. reverse path forwarding

- applicable even if routers do not have a knowledge of spanning trees
- when a packet arrives, the router checks if it arrived on the line that is normally used for sending packets to the source (the shortest path)

4.4 Packetizing and Fragmenting

- **Packetizing**: encapsulates packets received from upper-layer protocols and makes new packets out of them; done by the IP protocol in the Internet model
- **Fragmenting**: a datagram can travel through different networks; each router decapsulates the IP datagram from the received frame, processes it, and then encapsulates it in another frame
 - the format and size of the
 - **received frame** depends on the protocol used by the physical network from which the frame has just arrived
 - **departing frame** depends on the protocol used by the physical network to which the frame is going

- **Other Issues** (of Internet that are not directly related to the duties of the network layer)
 - **Address Resolution**: the network layer provides only host-to-host addressing whereas the data link layer needs physical addresses for node-to-node delivery; these addresses must be mapped; details later in the Address Resolution Protocol
 - **Multicasting** and **Routing Protocols** - covered earlier



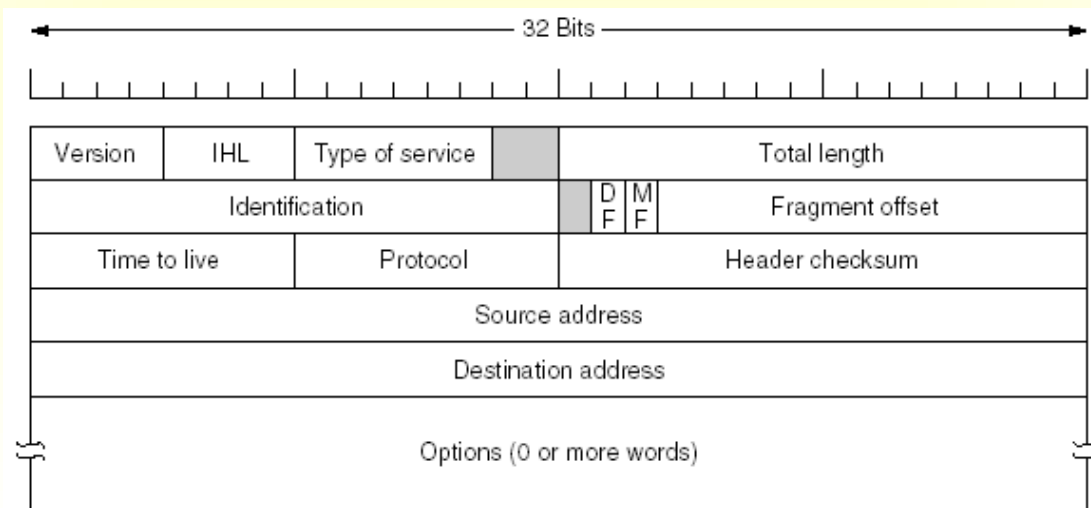
The Network Layer in the Internet

- there are 5 network layer protocols in the TCP/IP protocol suite (**IP**, **ARP**, **RARP** (**DHCP**), **ICMP**, and **IGMP**)
- the main protocol is **IP**, the glue that holds the whole Internet together and responsible for host-to-host delivery
- it needs the services of other protocols
 - **ARP** (Address Resolution Protocol) - maps an IP address to a MAC address (of the next hop)
 - **RARP** (Reverse ARP) - maps a MAC address to an IP address; usually used in some situations such as when a diskless host is booted; it gets the binary image of its operating system from a remote file server but does not know its IP address; obsolete, replaced by **DHCP**- Dynamic Host Configuration Protocol
 - **ICMP** (Internet Control Message Protocol) - to handle unusual situations such as the occurrence of an error
 - **IGMP** (Internet Group Management Protocol) - for multicasting since IP is designed for unicast delivery; you can read about it

4.4 IP v4 and Subnetting

- unlike the older ones, it was designed with internetworking in mind
- it is **unreliable** and **connectionless** datagram protocol - a **best effort** delivery service (no **error control** or **flow control**); it only detects errors and discards the packet if it is corrupted
- if reliability is important, it must be paired with a reliable protocol such as TCP (at the transport layer); like the post office, get your important mails registered
- an IP datagram consists of a header part and a text part
- the header has a 20-byte fixed part and a variable length optional part
- it is transmitted in **big endian** order: from left to right, with the high-order bit of the Version field going first (SPARC is big endian; Pentium is **little endian**); software conversion is required on little endian machines

- **Version** (4 bits): keeps track of which version of the protocol the datagram belongs (for future transitions between new and old versions); currently 4
- **IHL** (4 bits): length of the header in 32-bit words (minimum value is 5, which applies when no options are present; maximum value is 15, which limits the header to 60 bytes, and thus the Options field to 40 bytes)
- **Type of service** (8 bits): designed to distinguish between different classes of service - for various combinations of reliability (e.g. file transfer) and speed (e.g. voice); mostly ignored by current routers



The IPv4 header

■ IP Addresses

- every host and router on the Internet has an IP address (its network number and host number)
- unique and universal for global communication (no 2 hosts have the same address, but a host can have 2 IP addresses)
- 32-bit in IPv4 written as 4 decimal numbers (128 in IPv6)
- with 32-bits there are 2^{32} (4,294,267,296) hosts; assumed too many initially; now not sufficient due to a number of reasons (mainly wastage)
- **Classful Addressing**
 - address space is divided into 5 classes: A, B, C, D, and E
 - the first few bits indicate the class of an address

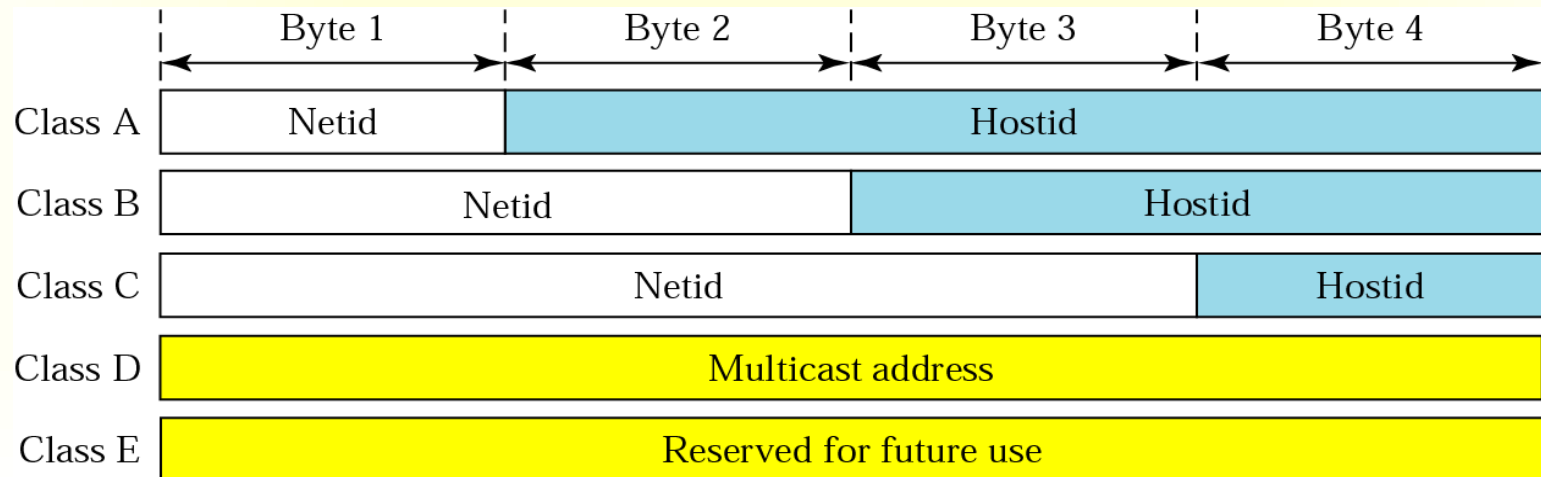
	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

in binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

in dotted-decimal notation

- addresses in classes A, B, and C are for **unicast** communication; a host needs to have at least one unicast address to be able to send and receive
- addresses in class D are for **multicast** communication (only as a destination); if a host belongs to a group or groups, it can have one or more multicast addresses
- addresses in class E are **reserved**; the idea was to use them for special purposes
- **Netid and Hostid**
 - addresses in classes A, B, and C are divided into netid (network) and hostid (host) of varying length from class to class



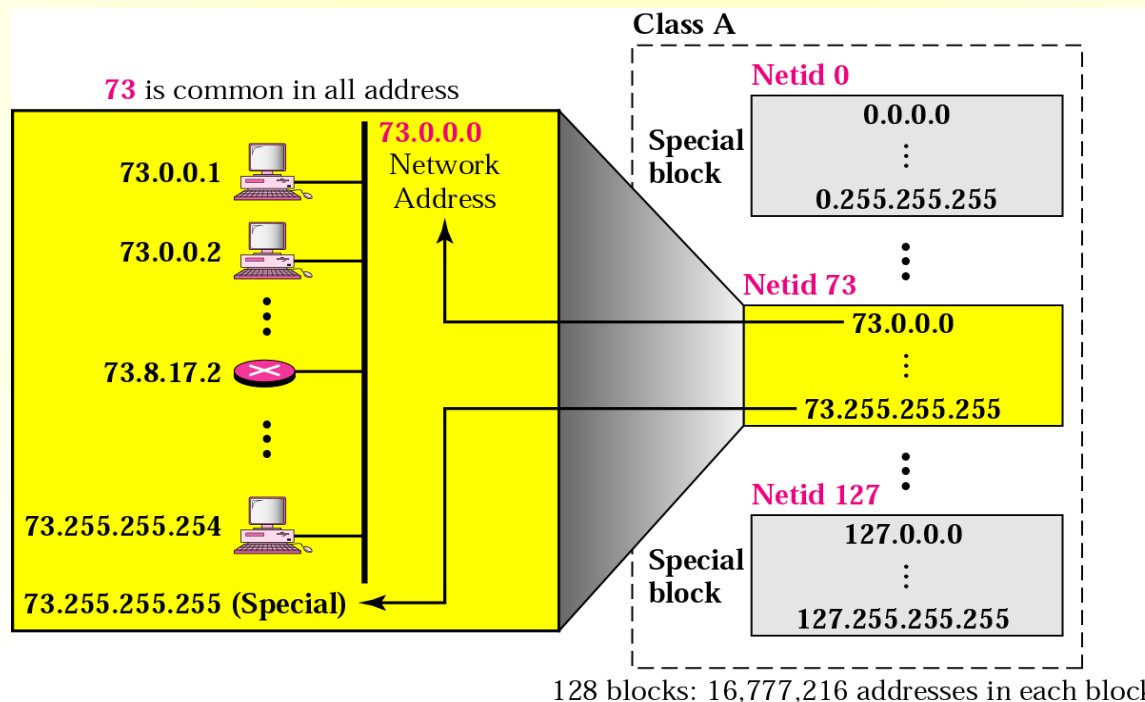
■ **Classes and Blocks**

- one problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size

■ **Class A**

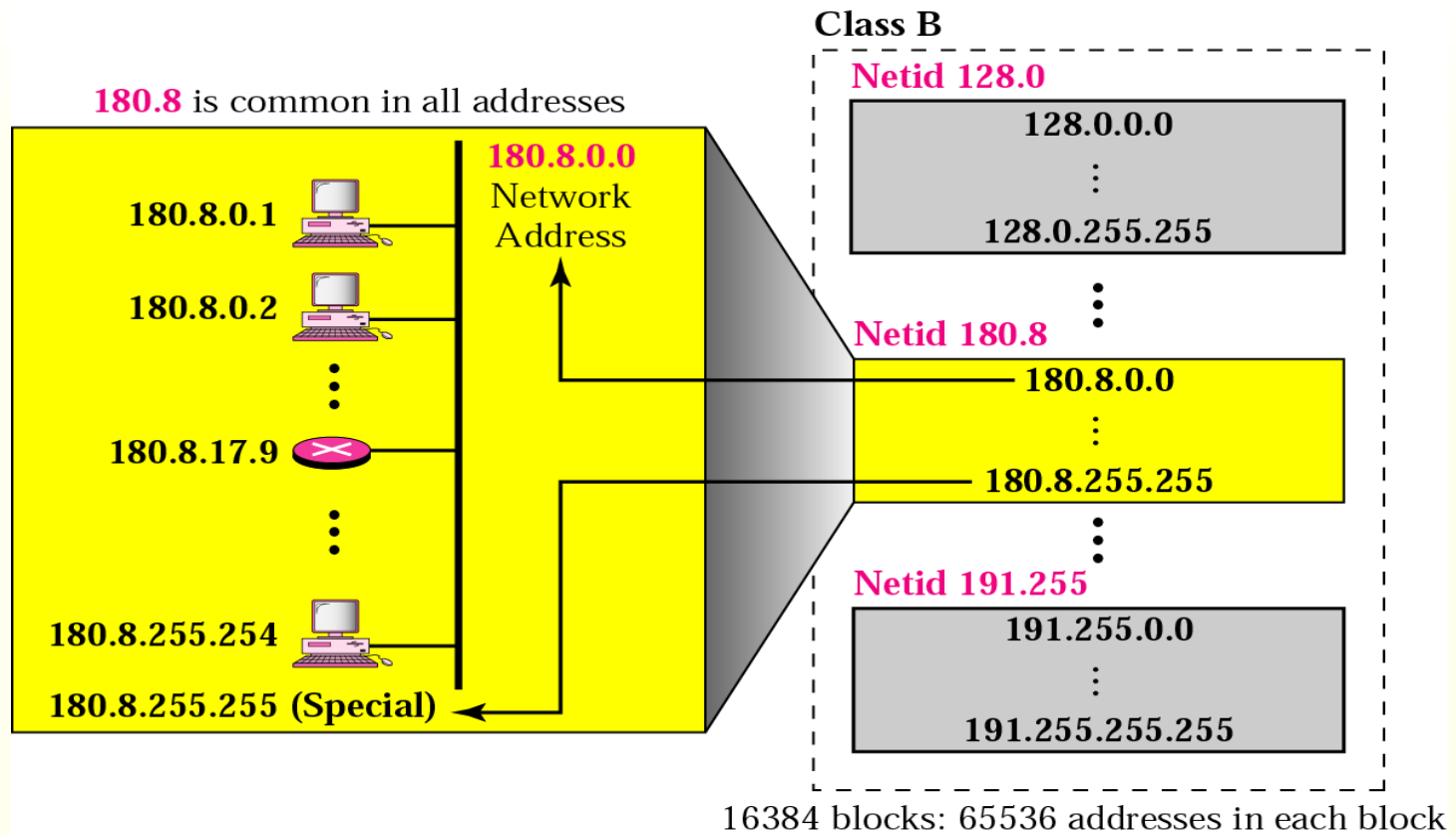
- divided into 128 blocks, each block having a different netid (the first bit of byte 1 is always 0)
- block 1: from **0.0.0.0** to **0.255.255.255** - netid **0**
- block 2: from **1.0.0.0** to **1.255.255.255** - netid **1**
- :
- last block: from **127.0.0.0** to **127.255.255.255** - netid **127**
- class A addresses were designed for large organizations (up to 16 million hosts 2^{24})

- e.g. organization X granted a block with netid 73
 - the first address (73.0.0.0) is used to identify the organization (called the **network address**)
 - the last address is reserved for a special purpose
 - total number of organizations that can be assigned class A addresses is 126 (128-2) because all 0's and 1's are reserved
 - the number of addresses in each block ($16,777,214 = 2^{24} - 2$) are larger than the needs of almost all organizations
 - **millions of class A addresses are wasted**



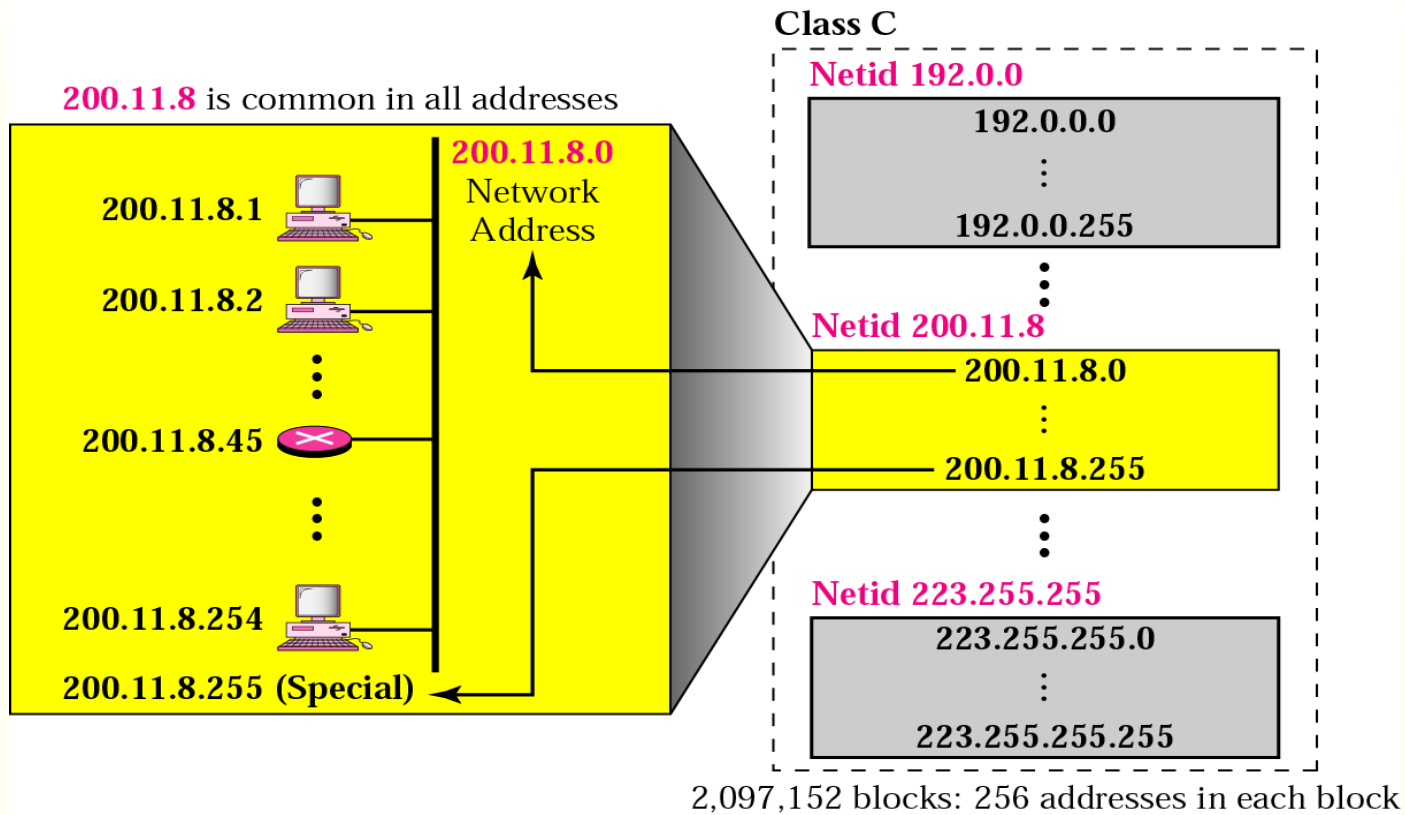
■ **Class B**

- divided into 16,384 (2^{14}) blocks, each block having a different netid (the first two bits of byte 1 are always 10)
- block 1: from **128.0.0.0** to **128.0.255.255** - netid **128.0**
- block 2: from **128.1.0.0** to **128.1.255.255** - netid **128.1**
- ■
- last block: from **191.255.0.0** to **191.255.255.255** - netid **191.255**
- 16 blocks are reserved for private addresses
- total number of organizations that can be assigned class B addresses is 16,368 ($16,384 - 16$)
- class B addresses were designed for midsize organizations (up to $65,534 = 2^{16} - 2$ hosts)
- the number of addresses in each block is larger than the needs of most midsize organizations
- **many class B addresses are wasted**

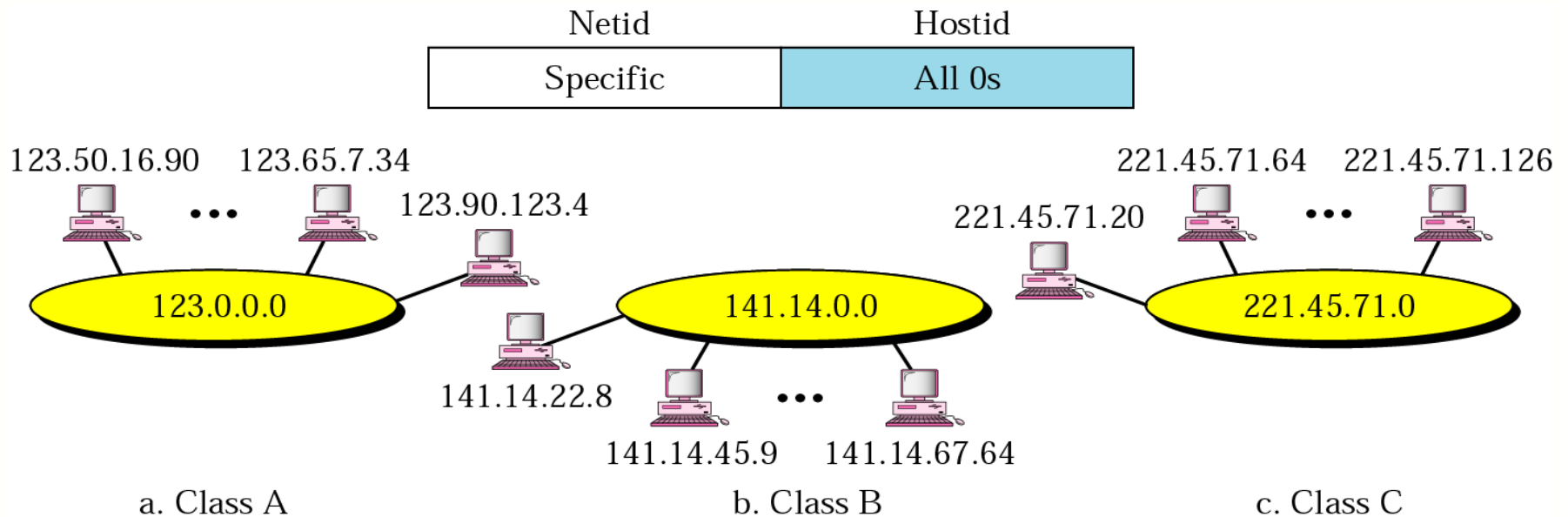


■ Class C

- divided into 2,097,152 (2^{21}) blocks, each block having a different netid (the first three bits of byte 1 are always 110)
- block 1: from **192.0.0.0** to **192.0.0.255** - netid **192.0.0**
- block 2: from **192.0.1.0** to **192.0.1.255** - netid **192.0.1**
·
·
- last block: from **223.255.255.0** to **223.255.255.255** - netid **223.255.255**
- 256 blocks are used for private addresses
- total number of organizations that can be assigned class C addresses is 2,096,902 (2,097,152-256)
- class C addresses were designed for small organizations (up to $254 = 2^8 - 2$ hosts)
- the number of addresses in each block is so limited that most organizations do not want a block in this class



- **Class D:** there is just one block of class D addresses; designed for multicasting
- **Class E:** there is just one block of class E addresses; designed for use as reserved addresses; for research by IETF (Internet Engineering Task Force) at <http://www.ietf.org>; it develops Internet standards
- **Network Address**
 - an address that defines the network itself
 - it cannot be assigned to a host
 - properties
 - all hostid bytes are 0s; it is different from a netid, a network address has both netid and hostid, with 0s for the hostid
 - it defines the network to the rest of the Internet; routing to a host is based on the network address
 - in classful addressing, the network address is the one that is assigned to the organization



- **network addresses are managed by a nonprofit organization to avoid conflicts - ICANN (Internet Corporation for Assigned Names and Numbers); it has delegated parts of the address space to various regional authorities, which then allocate IP addresses to ISPs and other companies**

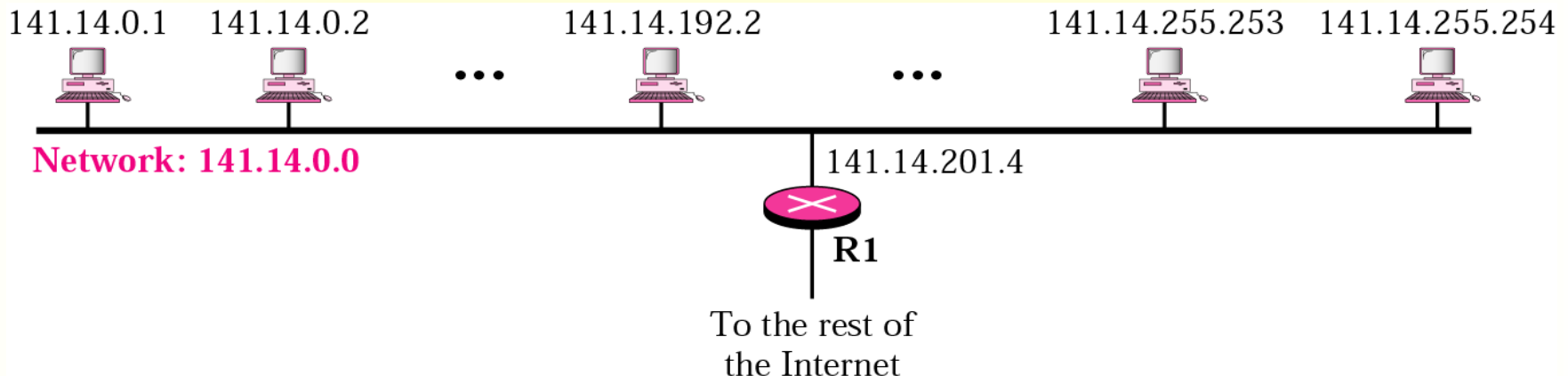
■ Special IP addresses

- an IP address with all 0s (hostid **and/or** netid) means this **network** or this **host**
- an IP address with all 1s (netid **and/or** hostid) means **all the hosts on the indicated network for broadcasting**

0 0																														This host			
0 0								...								0 0								Host								A host on this network	
1 1																														Broadcast on the local network			
Network								1 1 1 1								...								1 1 1 1								Broadcast on a distant network	

■ Subnetting

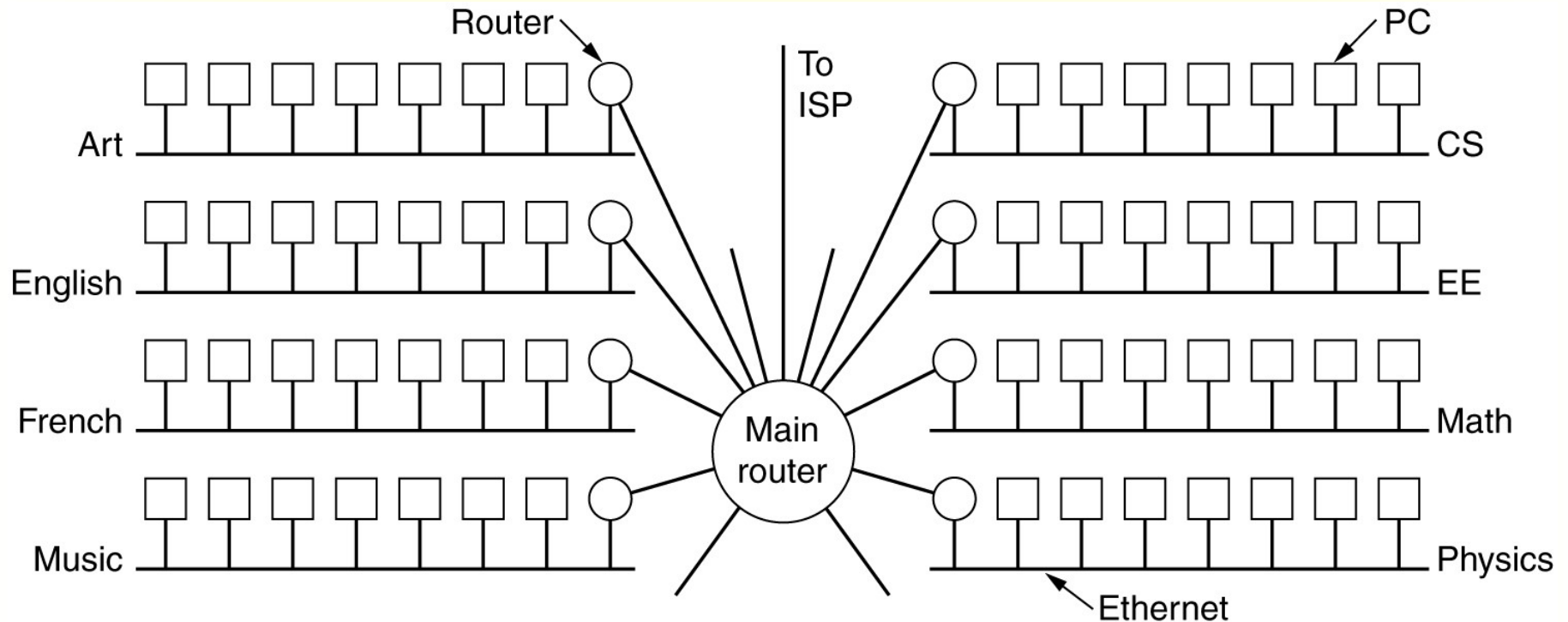
- the netid is used by routers to route a packet destined for a network
- IP addresses are designed with two levels of hierarchy (**site** and **host**)



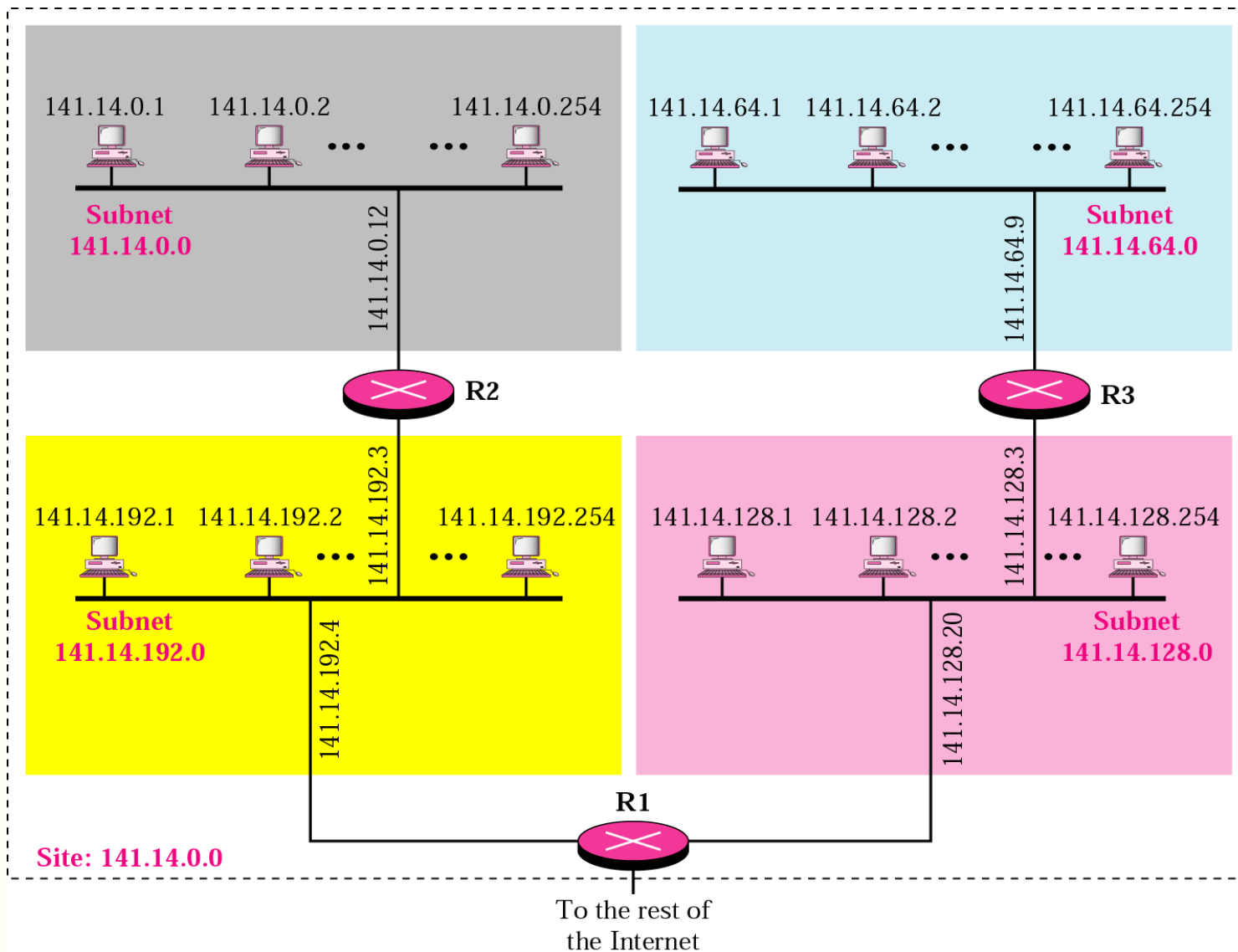
a network with two levels of hierarchy

- all the hosts in a network must have the same netid
- problem: does not allow dividing of a network into logical groups (such as Departments in a University)

- **solution 1: get netids for all groups; impossible**
- **solution 2: allow a network to be split into several parts for internal use but still act like a single network to the outside world (three levels of hierarchy: **site**, **subnet**, **host**)**

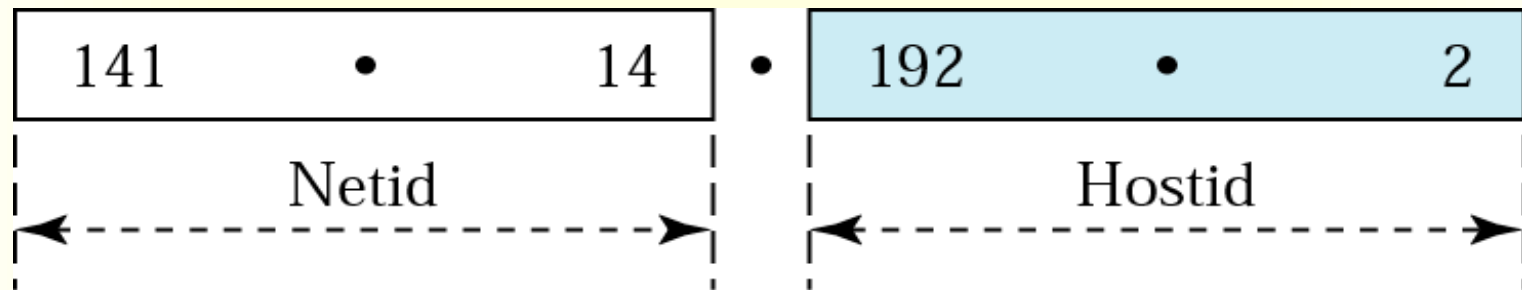


a campus network consisting of LANs for various departments

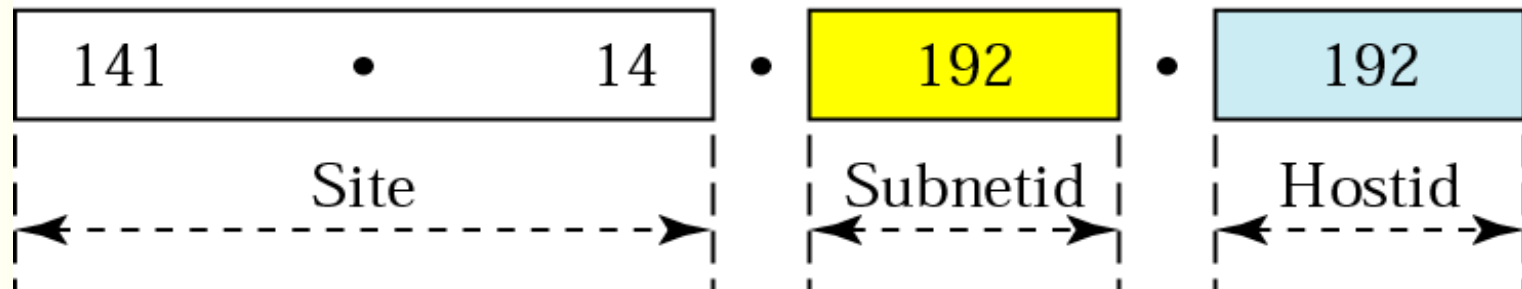


a network with three levels of hierarchy

- the parts of the network are called **subnetworks** or **subnets** (different from our previous definition of a subnet as the set of all routers and communication lines in a network)
- routing of a datagram now involves three steps: delivery to the site, delivery to the subnetwork, and delivery to to the host



a. Without subnetting



b. With subnetting

addresses in a network with and without subnetting

- the IP hostid is divided into subnetid and hostid
- e.g., for class B addresses subnetid could be 6 bits ($2^6=64$ groups or subnets) and hostid could be 10 bits ($2^{10}-2 = 1022$ hosts in each subnet)
- this could be changed later if required (say if there are more subnets) by the organization itself without any authorization, say by ICANN
- note: the first and last host numbers in a subnet are reserved; the first to identify the subnet itself; the last to broadcast within the subnet

■ Masking

- how can a router find the network address or subnet address?
- a 32-bit number called the **mask** is used; routers outside the organization use a **default mask**; routers inside the organization use a **subnet mask**
- **default mask**: a 32-bit binary number that gives the network address when ANDed with an address in the block; the number of 1s is the same as the number of bits in the netid (8 for A, 16 for B, 24 for C); the rest are all 0
- **slash notation**: an alternative mask notation, a slash followed by the number of 1s

Class	In Binary	In Dotted-Decimal	Using Slash
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

- e.g. a router outside the organization receives a packet with destination address 190.240.7.91; how does it find the network address to route the packet?
- the router follows three steps:
 1. it looks the first byte of the address to find the class; it is class B
 2. the default mask for class B is 255.255.0.0; it ANDs this mask with the address to get 190.240.0.0
 3. it looks in its routing table to find out how to route the packet to this destination

- **subnet mask**: change some of the leftmost 0s in the default mask to 1s to make a subnet mask

	255.255.0.0			
Default Mask	11111111 11111111		00000000 00000000	
			16	
	255.255.224.0			
Subnet Mask	11111111 11111111		111	00000 00000000
			3	13

- the number of subnets is determined by the number of 1s (2^n)
- conversely, if the number of subnets is n, the number of extra 1s is $\log_2 n$

- e.g. a router inside the organization receives a packet with destination address 190.240.33.91; how does it find the subnetwork address to route the packet?
- the router follows three steps:
 1. the router must know the mask; assume it is /19 as in the above figure
 2. it applies the mask to the address; the subnet address is 190.240.32.0
 3. it looks in its routing table to find out how to route the packet to this destination

■ Classless Addressing

- there is too much wastage of IP addresses in classful addressing creating shortage of IP addresses
- classless addressing was announced in 1996
- the idea of **variable-length blocks** that belong to no class was introduced to allocate the **remaining** IP addresses
- blocks of 2 addresses, 4 addresses, 64 addresses, ... can be assigned to an organization based on size
- the number of addresses in a block must be a power of 2
- the whole address space (2^{32} addresses) is divided into blocks of different sizes
- the beginning address must be **evenly** divisible by the number of addresses; if a site needs, say, 2000 addresses it is given a block of 2048 addresses on a 2048 address boundary
- we can still use **subnetting** if there is a need
- an organization is given the **beginning address** of the block and a **mask** (in slash notation)

- **classless addressing alleviates the depletion of addresses, but we now need a new method of forwarding packets, called **CIDR (Classless InterDomain Routing)****
- **there is now a single routing table for all networks consisting of an array of (IP address, subnet mask, outgoing line) triples**
- **when a packet comes in, its destination IP address is first extracted; the routing table is then scanned entry by entry, masking the destination address and comparing it to the table entry looking for a match**
- **it is possible that multiple entries (with different subnet mask lengths) match, in which case the longest mask is used; if there is a match for a /20 mask and a /24 mask, the /24 entry is used**

University	First address	Last address	How many	Written as
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

- consider the previous example; the table entries (in binary) for the three universities are

Base (First) Address	Mask
C: 11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000
E: 11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000
O: 11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000

- now consider what happens when a packet comes that is addressed to 194.24.17.4, which in binary is represented as
11000010 00011000 00010001 00000100
- AND it with each mask and compare it with the base address
- first it is ANDed with the Cambridge mask to get
11000010 00011000 000**1**0000 00000000; does not match the Cambridge base address
- then the original address is ANDed with the Edinburgh mask to get
11000010 00011000 000**10**000 00000000; does not match the Edinburgh base address

- Oxford is tried next, yielding
11000010 00011000 00010000 00000000; it matches the Oxford
base
- if no longer matches are found farther down the table, the
Oxford entry is used and the packet is sent along the line named
in it
- Read about NAT - Network Address Translation

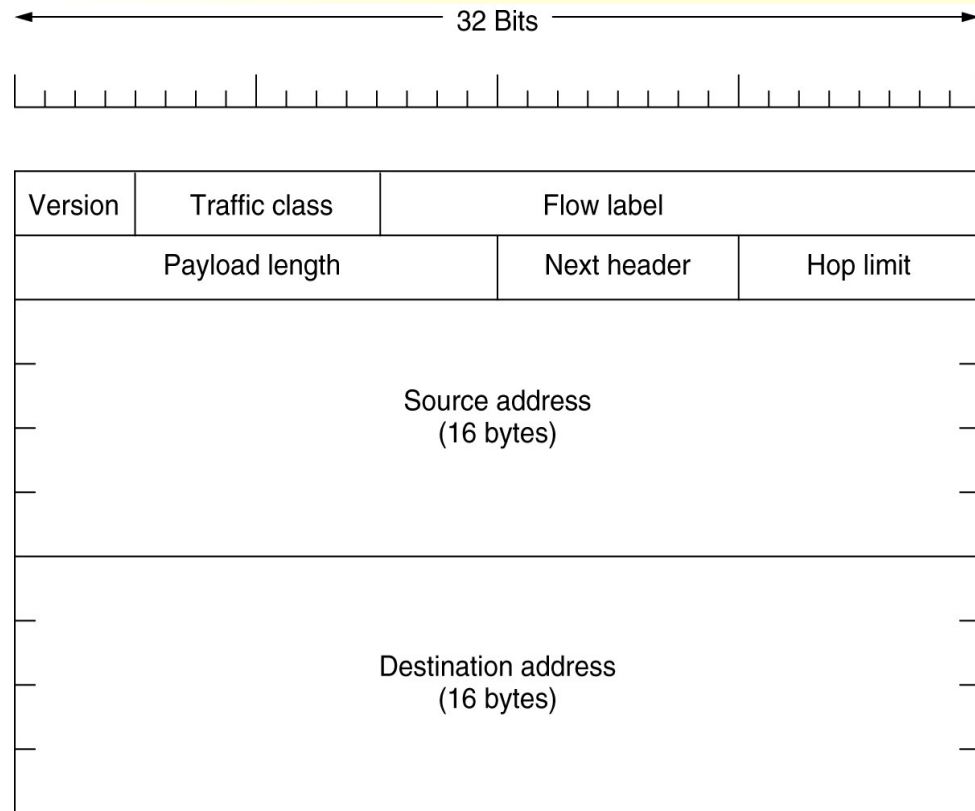
■ IPv5

- is (was) there IPv5?
- yes, it was an experimental real-time stream protocol that was never widely used
- was designed to coexist with IPv4, not a replacement

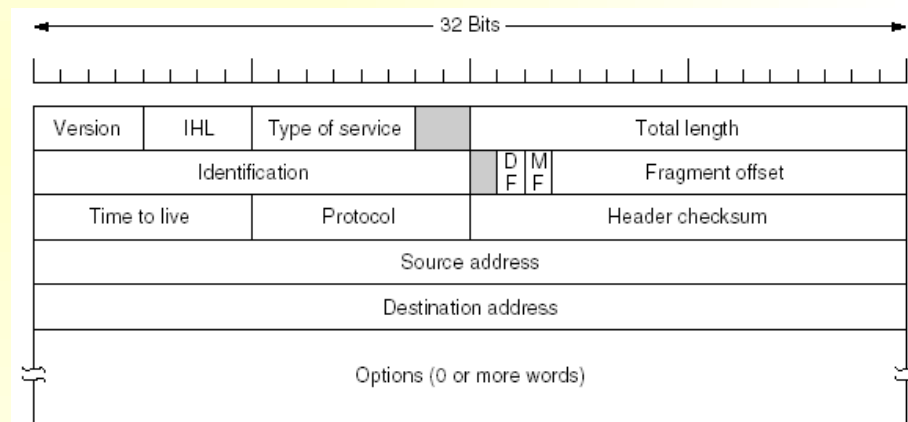
4.5 IPv6

- the major goals are
 - **Larger address space**: to support billions of hosts, even with inefficient address space allocation (2^{128} of them - if the entire earth, land and water, were covered with computers, IPv6 would allow 7×10^{23} IP addresses per square meter - Tannenbaum)
 - **Better header format**: to simplify and speed-up the routing process
 - **Better security**: better encryption and authentication options
 - **Support for resource allocation**: to deal with the special requirements of real-time data (audio and video)
 - **Allowance for extension**: to allow the extension of the protocol if required by new technologies or applications in the future
 - **Coexistence**: permit the old and new protocols to coexist for years

- IPv6 header has 8 fields (40 bytes long)
- **Version** (4 bits): is always 6 for IPv6 (and 4 for IPv4)
- **Traffic class** (4 bits): used to distinguish between packets with different real-time delivery requirements
- **Flow label** (24 bits): to provide special handling for a particular flow of data; for example, to reserve bandwidth because of some delay requirements



The IPv6 header



The IPv4 header

- **IPv6 addresses notation**

- **hexadecimal colon notation**: written as eight groups (each 2 bytes) of four hexadecimal digits with colons between the groups

8000:0000:0000:0000:0123:4567:89AB:CDEF

- many addresses usually will have many zeros
- three **abbreviation** methods
 - leading zeros (not the trailing ones) within a group can be omitted; 0123 can be written as 123
 - one or more groups of 16 zero bits can be replaced by a pair of colons; the above address now becomes

8000::123:4567:89AB:CDEF

- it is allowed only once per address (if there are two runs of zero sections, only one of them can be abbreviated)
- IPv4 addresses can be written as a pair of colons and an old dotted decimal number, for example

::192.31.20.46

■ Categories of Addresses

■ IPv6 defines 3 types

■ Unicast, Multicast

- **Anycast**: defines a group of computers with addresses that have the same prefix; such a packet must be delivered to **exactly** one of the members of the group - the closest or the most easily accessible

■ transition from IPv4 to IPv6

- it can not happen suddenly; smooth transition is required
- three strategies designed: **dual stack**, **tunneling**, and **header translation**

4.6 Address Resolution

- the data link layer hardware does not understand IP addresses
- e.g., every Ethernet board comes with a 48-bit Ethernet address (issued by a central authority to avoid conflicts); such boards send and receive frames based on 48-bit Ethernet addresses; they know nothing about 32-bit IP addresses
- a MAC address is a local address
- delivery of a packet to a host or a router requires two levels of addressing: IP and MAC
- ➡ a mapping of IP addresses to MAC addresses is required
- two types of mapping: **static** and **dynamic**

■ **Static Mapping**

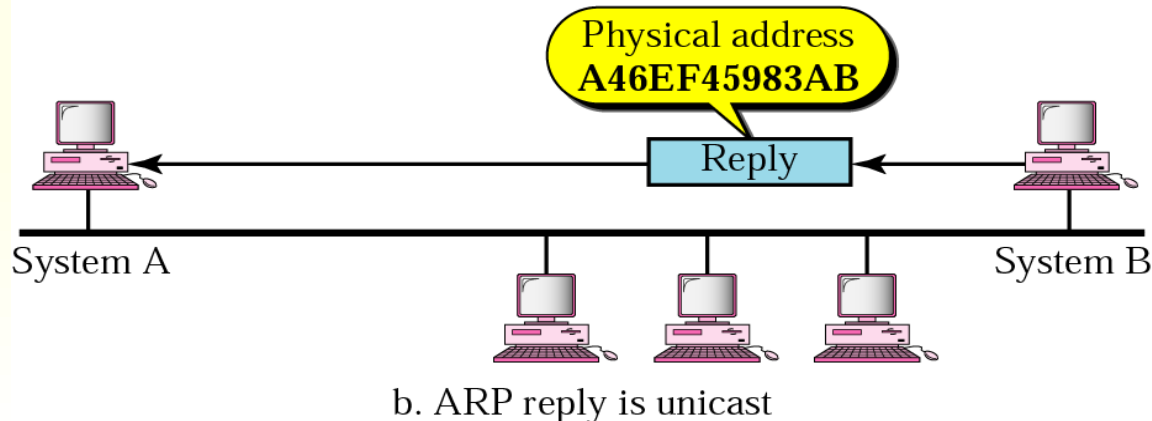
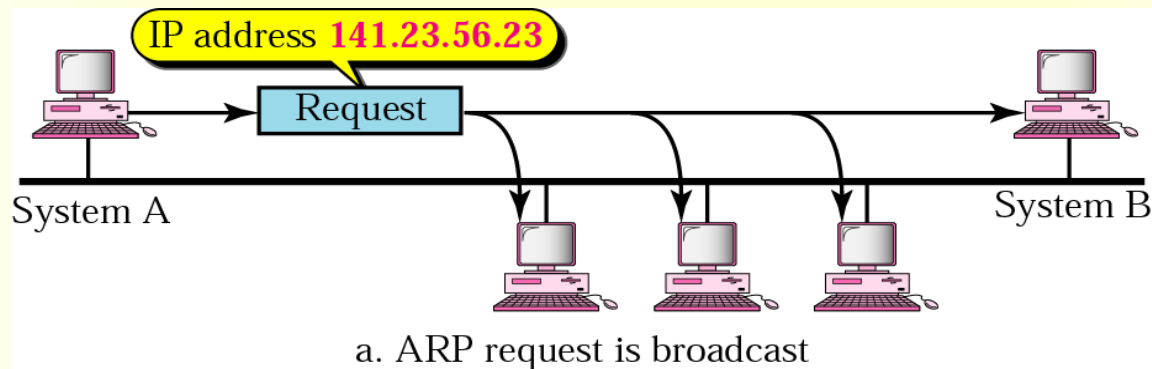
- create a table that associates an IP address with a MAC address, stored in each machine on a network
- network performance is degraded to update the table periodically (a machine could change its network card; in some LANs such as LocalTalk of apple, the MAC address changes every time the computer is turned on; a mobile computer can move from one physical network to another)

■ **Dynamic Mapping**

- each time a machine knows one of the two addresses, it can use a protocol to find the other one
- two protocols: **ARP** (maps an IP address to a MAC address) and **RARP** (maps a MAC address to an IP address; obsolete, replaced by **DHCP**- Dynamic Host Configuration Protocol)

■ ARP (Address Resolution Protocol)

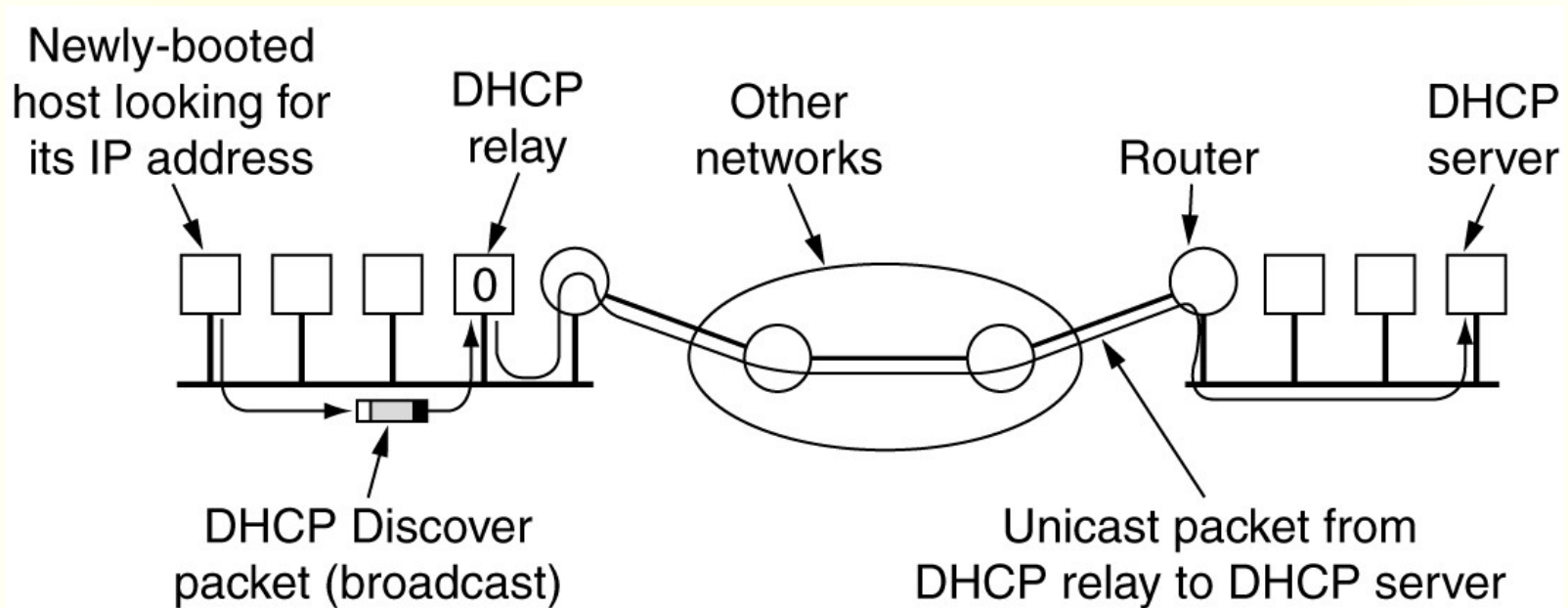
- a host or a router looking for a MAC address broadcasts an ARP query packet - it includes the MAC (physical) and IP addresses of the sender and the IP address of the receiver
- only the intended recipient sends back an ARP response packet (it contains the recipient's IP and physical addresses); it is unicast



- **DHCP (Dynamic Host Configuration Protocol)**
 - each computer connected to the Internet must have the following information
 - its IP address
 - its subnet mask
 - the IP address of a router
 - the IP address of a name server
 - such information can be saved in a configuration file of the client; but what if
 - it is diskless or
 - it is booting for the first time or
 - it has moved to another subnet
 - DHCP is a client-server program for assigning network addresses
 - IP addresses, default routers
 - it is an extension of **BOOTP** (that replaced **RARP**) that maps IP addresses to Ethernet addresses; but it requires manual configuration of the table by an administrator when a new host is added - static

- **DHCP allows both manual and automatic IP address assignment**
- **a DHCP server has two databases**
 - **one that statically binds physical addresses to IP addresses**
 - **the second holds a list of unassigned IP addresses that makes DHCP dynamic**
- **computers contact the DHCP server at boot-up time**
 - **the server checks its static database; if there is an entry that permanent address is returned**
 - **otherwise it dynamically assigns an IP address for a fixed period of time**
 - **i.e., addresses are “leased” for some time; when the time expires the client must request a renewal**
- **this allows sharing of address space**
 - **results in more efficient use of address space**

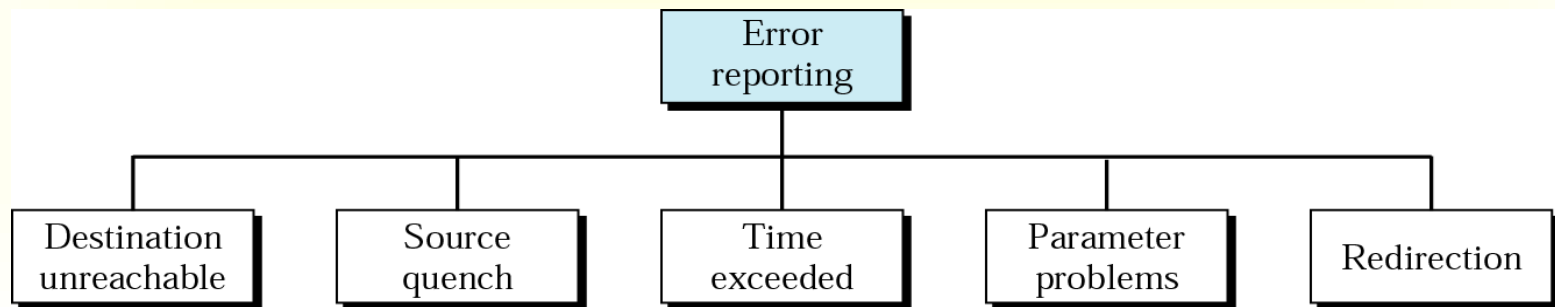
- a **DHCP relay agent** is needed on each LAN since the DHCP server may not be reachable by broadcasting



Operation of DHCP

■ ICMP (Internet Control Message Protocol)

- IP has no **error reporting** and error correcting mechanisms; if something goes wrong
- it lacks a mechanism for **host** and **management queries** (is a router or a host alive?)
- ICMP is a companion to the IP designed to compensate these two deficiencies
- two type of ICMP messages: **error-reporting** and **query** messages
- **Error Reporting**
 - always reports error messages to the original source (the only information included in the datagram is the source and destination addresses)



- **Destination unreachable**: when a router cannot locate the destination or when a packet with the DF bit cannot be delivered because a “small-packet” network stands in the way; the datagram is discarded
- **Source quench**: to slow down a source since IP is connectionless and lacks flow control and congestion control; rarely used because when congestion occurs, these packets tend to aggravate it; congestion control in the Internet is handled in the transport layer
- **Time exceeded**: sent when a packet is dropped because its counter (**Time to live**) has reached zero (a symptom that packets are looping) or when all fragments that make up a message do not arrive at the destination host within a certain time limit
- **Parameter problem**: indicates that an illegal value has been detected in a header field
- **Redirection**: when a router notices that a packet seems to be routed wrong